

Objectives

To analyze, select, and implement an appropriate algorithmic framework to predict student dropout risk across large datasets. The focus is on evaluating the trade-offs between complex machine learning classifiers and a distributed, rule-based heuristic scoring model to build an interpretable, highly scalable predictive baseline.

Tasks Completed

1. Reviewed established predictive algorithms including logistic regression, decision trees, and gradient boosting.
2. Evaluated computational trade-offs against the pipeline's 1,000,000-row volume.
3. Selected a rule-based heuristic model, assigning positive weights to risk factors and negative weights to protective factors.
4. Programmed percentile-based cutoffs (33rd and 66th) to segment scores into Low, Medium, and High Risk tiers.
5. Configured the pipeline to export the algorithm's classified output into a finalized CSV file tailored for subsequent machine learning phases.

Tasks to be Completed

N/A

Results and Analysis

1. The chosen rule-based algorithm scaled perfectly across Spark's distributed architecture, processing all 1,000,000 rows without memory failure.
2. The model provided a transparent look at risk drivers, demonstrating that psychological exhaustion from burnout (+1.5) and stress (+1.4) impacts dropout trajectories more than standard exam pressure (+1.0).
3. By applying percentile cutoffs to a normal distribution, the algorithm successfully isolated the 1.5% high-risk minority tier. This generated pre-labeled data rows, creating a clean foundation for training complex machine learning models.

Challenges and Solutions

Challenge: Training compute-heavy machine learning classifiers directly on 1,000,000 raw records causes severe single-node bottlenecks.

Solution: Used PySpark RDDs to distribute the initial heuristic scoring rules across parallelized chunks, exporting a structured CSV file optimized for subsequent machine learning models.

Conclusion

The lab successfully achieved its objectives by selecting a distributed rule-based heuristic algorithm to serve as the initial classification layer. This choice avoided localized processing bottlenecks, clarified the primary psychological catalysts of attrition, and successfully isolated the high-risk student population. Ultimately, choosing this transparent model demonstrates that structured data scoring is an essential prerequisite for deploying reliable machine learning frameworks.